EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

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Detailed mark scheme

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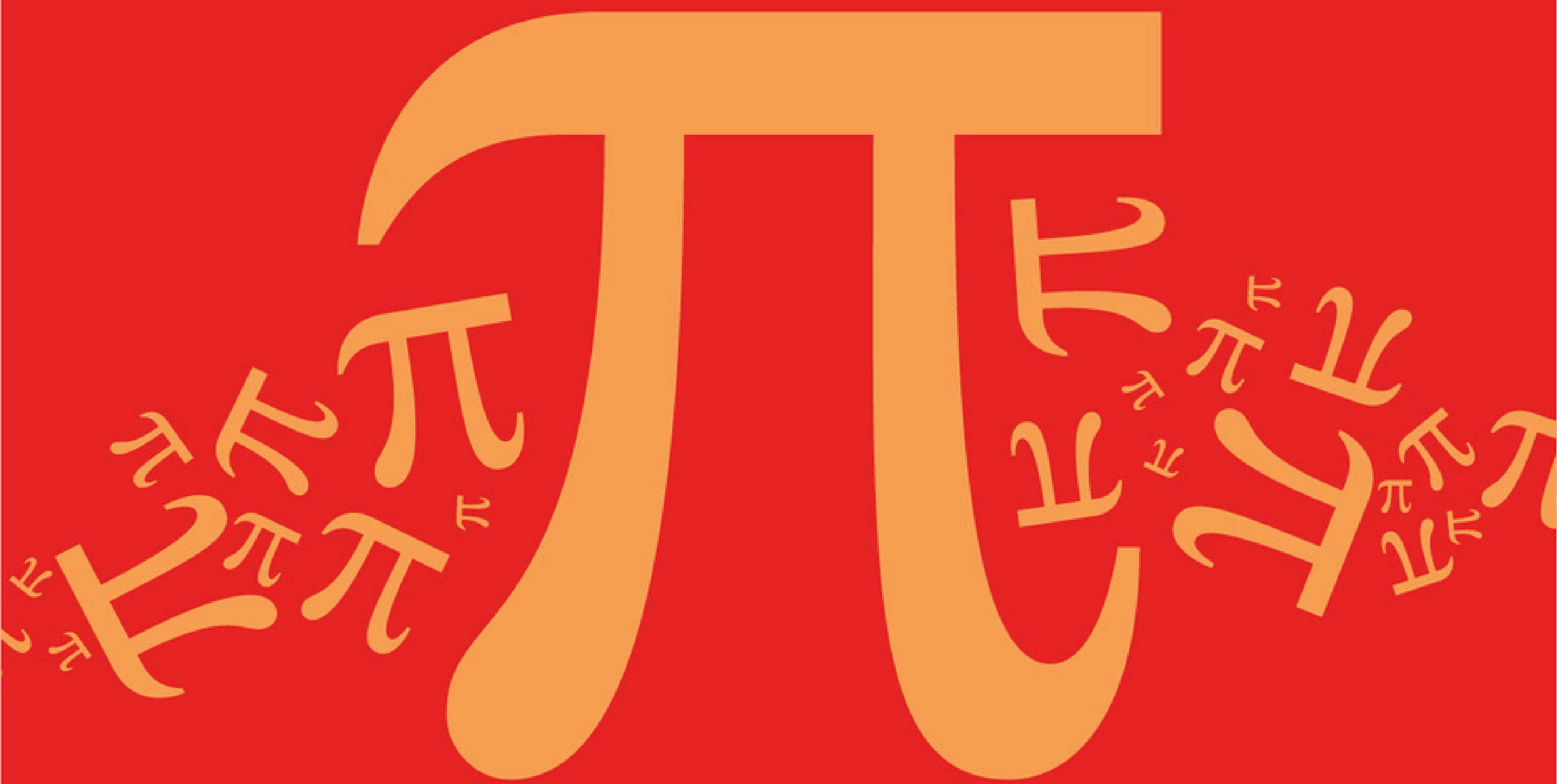
AQA A Level Further Mathematics (7367)



Topic DB: Networks

Sub topic DB3: Route Inspection Problems

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic **DB: Networks**

Sub topic **DB3: Route Inspection Problems**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Gives a plausible reason, in the context of the question, as to what *** could mean, such as no cycle path exists	AO2.4	E1	A cable cannot be laid between these two districts as there may be a river or housing estate in the way
(b)	Identifies the problem as a route inspection problem by noting that *B*, *G*, *H* and *O* are odd-degree nodes. (PI)	AO3.1b	M1	odd nodes: *B*, *G*, *H* and *O*
	Finds the shortest distance between each pair of odd nodes (at least four correct) OR determines correctly values for the *** entries in the table, with clear reasoning.	AO1.1b	M1	B-G: 2.5; B-H: 5.6; B-O: 4.3; G-H: 3.1; G-O: 6.8; H-O: 6.7. OR B-H: 5.6 (via G) and G-O: 6.8 (via B)
	Determines correctly the minimum total distance the engineer will cover during the journey. CAO	AO1.1a	A1	Minimum pair of shortest distances is B-O and G-H, with a minimum extra distance of 7.4 miles 37.2 + 7.4 = 44.6 miles
	Determines the minimum time in minutes that the engineer could complete the journey in by using ‘their’ minimum total distance and the average speed of 12 miles per hour.	AO3.2a	A1F	(44.6 / 12) × 60 = 223 minutes.
				Total 5 marks

Q2.

(a) (Odds B, D, F, H)

$BD + FH = 37.2$

$BF + DH = 38.4$

These 3 pairs of odds stated

M1

$BH + DF = 40$

3 correct totals, 2 correct totals

A2,1

Length 118 + 37.2

118 + their 'smallest' PI by their final answer

m1

= 155.2

CSO, including 3 correct totals.

A1

5

(b) (i) E twice

B1

(ii) I twice

B1

1

[7]

Q3.

(a) (Odds A, C, L, G)

$AC + LG = 27$

$AL + CG = 26$

$AG + CL = 30$

These 3 sets of pairs stated

M1

One mark for each correct total

A1 x 3

Min 134 + 26

134 + their min of 3 totals.

m1

= 160

Must have scored first 5 marks.

A1

If M0 scored, then 160 scores SC2

CSO

6

(b) 4

B1

1

[7]

Q4.

(a)

$$\left.\begin{aligned} CE + KH &= (35 + 24) = 59 \\ CK + EH &= (25 + 40) = 65 \\ CH + EK &= (25 + 30) = 55 \end{aligned}\right\}$$

These 3 correct sets of pairs

M1

3 correct totals, 2 correct totals

A2,1

Total = 224 + 55 PI by their ‘279’

224 + their smallest of three pair totals

M1

= 279

CSO including totals seen

A1

5

(b) 3

B1

1

[6]

Q5.

(a)

$$\begin{aligned} BD+FH &= \\ BF+DH &= \\ BH+DF &= \end{aligned} \left[\begin{array}{c} 210+210 \\ 200+180 \\ 260+340 \end{array}\right] \begin{array}{l} = 420 \\ = 380 \\ = 600 \end{array}$$

These 3 sets of pairs

M1

3 correct totals, 2 correct totals

A2,1

(MIN) = 2430 + 380

2430 + their smallest of three pair totals

m1

= 2810

CSO

A1

5

(b) 2430 + 340 (DF) = 2770

2430 + their DF

B1F

1

(c) (i) 2430 + 180 (DH) = 2610

2430 + their min (must have scored M1)

B1F

1

(ii) B, F only

B1

1

[8]

Q6.

(a) A vertex / vertices of odd order (A, B, G, H) OE

Condone statement of non-Eulerian graph

E1

1

(b)

$$\left.\begin{aligned} AB + GH &= (180 + 165) = 345 \\ AG + BH &= (90 + 210) = 300 \\ AH + BG &= (150 + 210) = 360 \end{aligned}\right\}$$

These 3 correct sets of pairs

M1

3 correct totals, 2 correct totals

A2,1

Dist 1215 + 300 PI

1215 + their smallest

M1

= 1515

CSO

A1

5

(c) (i) 3

B1 1

(ii) 2

B1 1

[8]

Q7.

(a) $AC + FD$ (= 14 + 18) = 32

These 3 correct sets of pairs, letters not numbers

M1

$AF + CD$ (= 10 + 26) = 36

3 correct totals, 2 correct totals

A2,1

$AD + CF$ (= 26 + 24) = 50

Condone 26 + 24 not evaluated if statement of "too big" OE

min = 150 + 32

150 + their smallest, PI

m1

= 182

A1cso 5

(b) Repeat FD

PI 182 – AC

M1

(= 150 + 18) = 168

168 unsupported scores 2/2

A1 2

(c) (i) Repeat AF

PI

M1

(= 150 + 10) = 160

160 unsupported scores 2/2

A1

2

(ii) (Start/finish) C and D

Must have both and only these

B1

1

[10]

Q8.

(a) (i)

AC	13
AE	14
EI	15
CD	16
CH	20
EF	21
FB	19
BG	19

Use of Prim's (not Kruskal's and not path); 6+ edges (no cycles); edges, not lengths or vertices, with first 2 edges correct

M1

8 edges

B1

CH *5th*

A1

EF *6th*

A1

All correct

A1

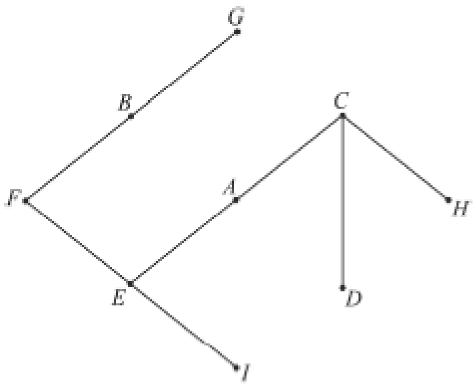
5

(ii) 137

B1

1

(iii)



6+ edges, no cycles

M1

Correct, including labelling

A1

2

(b) (Odds) B, C, D, E

PI CAO

E1

$BC + DE$ = 22 + 18 (or 40)
$BD + CE$ = 38 + 27 (or 65)
$BE + CD$ = 22 + 16 (or 38)

3 correct sets of pairs (lettered)

M1

3 correct sets of numbers; 2 correct sets of numbers

A2; 1

min = 307 + 38

PI 307 plus their shortest

A1F

= 345

B1

6

SC:
345 with no M mark scored scores 2/last 5
Route without 345 scores 0/last 5

[14]

Q9.

(a) Odd vertices A, B, C, M

PI, CAO

E1

$AB + CM$ = 25 + 48 or 73
$AC + BM$ = 24 + 49 or 73
$AM + BC$ = 47 + 23 or 70

3 correct sets of lettered pairs of candidate's vertices

M1

3 correct, 2 correct

A2, 1

Min = 384 + 70

PI, 384 plus their shortest

A1F

= 454

B1

SC
454 with no working, or 454 with route 2/6
Route without 454 0/6

6

(b) 4

B1

1

[7]

Q10.

Consider A, D, K, H

PI

B1

$AD + KH$ = 27 + 30 = 57

M1

$AH + DK$ = 20 + 20 = 40

A2,1,0

$AK + DH$ = 46 + 40 = 86
Total: 308 + 40 = 348

B1

[5]

Q11.

(a) Odds A, B, C, D

PI (but A, B, C, D must be mentioned)

M1

Considering 3 sets of pairings of odd vertices, eg AB with CD etc

m1

$$\left.\begin{aligned} AB + CD &= 270 + 270 = 540 \\ AC + BD &= 290 + 290 = 580 \\ AD + BC &= 260 + 270 = 530 \end{aligned}\right\}$$

A1 for 2 correct, A2 for all correct

A2,1,0

Repeat AD, BC

Follow through their shortest pairing
PI by adding 530 to 1920
*Or $AEHD$ or $DHEA$ **and** $BFGC$ or $CGFB$ listed in any route*

A1F

(Length = 1920 + 530 =) 2450 (metres)

B1

6

(b) Repeats BC

PI by $BFGC$ or $CGFB$ listed in a complete route or adding 270 / subtracting 260

E1

(Length = 1920 + 270 =) 2190 (metres)

2450 – 260 = 2190
(2190 with no evidence scores E0B1)

B1

2

(c) (i) Min. repeat AD

PI by $AEHD$ or $DHEA$ listed in a complete route or adding 260 / subtracting 270

E1

(Length = 1920 + 260 =) 2180 (metres)

2450 – 270 = 2180
(2180 with no evidence scores E0B1)

B1

2

(ii) B, C

Condone start at B, finish at C (or reverse)

B1

1

[11]

Q12.

(a) Odd vertices (at 4 B, C, I)

E1

1

(b) $AB + CI$ = 100 + 440 = 540

M1

$AC + BI$ = 150 + 450 = 600

A2,1,0

$AI + BC$ = 380 + 120 = 500

Repeat $AI + BC$

May be implied

E1

Distance 2090 + 500 = 2590

B1

5

(c) Route with

(3A), 2B, 2C, 3D, 2E, 2F, 3G, 1H, 2I, 1J

(16 → 21)

M1

= 18

A1

2

[8]

Q13.

(a) A, C, D, F odd nodes

May be implied

B1

$AC + DF = 18 + 22 = 40$

M1

$AD + CF = 32 + 30 = 62$

A2,1,0

$AF + CD = 12 + 30 = 42$
Repeat $AC + DF$

May be implied

B1

Total 164 + 40 = 204

B1

6

(b) Start / finish A/C
$\therefore$ Repeat DF

Or subtract AC

B1

Total 164 + 22 =186

B1

2

(c) (i) Shortest pair AF

B1

Distance = 164 +12 = 176

B1

2

(ii) Start / Finish at C/D

May be listed in a route

B1

1

[11]

Q14.

(a) Odd vertices (*ADFI*)

E1

1

(b) $AD + FI = 14 + 14 = 28$

M1

$AF + DI = 14 + 13 = 27$

$AI + DF = 11 + 17 = 28$

A2,1,0

$\therefore$ Repeat $AF + DI$

may be implied

E1

Distance = 87 + 27 = 114

B1

Route with

$3A, 1B, 2C, 2D, 3E, 2F, 1G, 1H, 2I$

17 vertices

B1

6

[7]

Q15.

Odd vertices N, A, I, X

M1

Min $NA + MIX = 65$

A1

No other pairings quicker

E1

$\therefore$ total time 790 (secs)

B1

[4]

Q16.

Odd vertices D and F

E1(*)

Repeat x or 13 (or DF)

(*) May be implied

B1(*)

$\therefore$ $2x + 82 = 100$

M1

$x = 9$

A1

4

[4]

Q17.

(a) (i) BC – 6

CD	–	8
CE	–	9
FH	–	9
GH	–	10
AC	–	11
GI	–	12
DF	–	14
AM	–	30
		109

B1

SCA

M1*

9 edges
**independent*
CE FH either order

M1*

Not DE

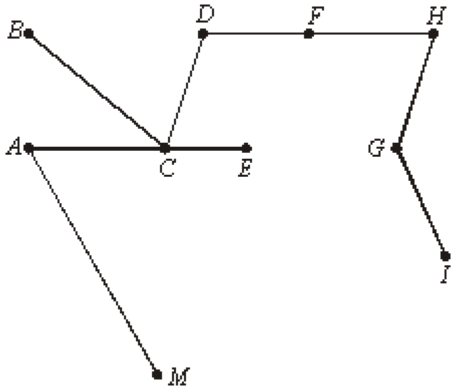
A1*

All correct
** Must have M2*

A1*

5

(ii)



9 edges

M1

A1
2

(b) Odd vertices
AB, *HI*

Min connect
AB = 17
HI = 22

B1

Reject *AH*, *BI* and *AI*, *BH*
or their values shown
(their total + their 39)

E1

Total 264 + 39
= 303 metres

M1

CAO
SC: 248 with no working 2/4
SC: MR with ABHIM: M1 repeat HB
A1 309

A1
4
[11]

Q18.

(a) (i) odd vertices

E1
1

(ii) C, D, E, F
CD + EF = 200 + 150 = 350

M1

CE + DF = 200 + 200 = 400

A2,1,0

CF +DE = 325 + 50 = 375
Repeat CGD& EF

B1

Tour with

M1

A2, B2,C2, D2,E2, F2, G3

A1

Distance $= 1500 + 350$
$= 1850$

B1F

7

(b) $1 \times 3 \times 5 = 15$

M1A1

2

[10]

Q19.

(a) (i) $(ABDK) = 3$

B1

1

(ii) $MST = (n - 1)$

M1

$= 15$

A1

2

(b) (i) Odd vertices A and P
Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

(33 + 3) = 36

B1

4

(ii) Repeated: AE, EL, LP

B1

1

(iii) E and L now odd

M1

Their(33) – 2 + 1 = 32

A1F

2

[10]

Q20.

(a) Odd vertices

E1

1

(b) BE $2x - 1$

BAE 13

$BCDE$ $x + 11$

B2, 1, 0

2

(c) $x = 10$ $\therefore$ repeat BAE

B1

Total = 53 + 13 = 66

for 53+

M1 A1

Route eg *BAEDCBEAB*

OE (Must have 9 letters)

B1

4

(d) BE $2x - 1 < 13$ $x < 7$

M1

$2x - 1 < x + 11$ $x < 12$

BAE $13 < 2x - 1$ $x < 7$

$13 < x + 11$ $x > 2$

M1

$BCDE$ $x + 11 < 2x - 1$ $x > 12$

$x + 11 < 13$ $x < 2$

SC: Substitute values
Either 1/5 (no explanation of values)
or complete table with x = 1 to x = 11 M2A2E1

A1

$\therefore$ not $BCDE$

or M1A1 $x \le 2 \Rightarrow BE < BCDE$
M1A1 $x > 2 \Rightarrow BAE < BCDE$
E1 $\therefore$ Not $BCDE$

E1

5

[12]

Q21.

(a) Graph 1: no – odd vertices (semi)

for considering vertices

M1B1

Graph 2: no – odd

B1

Graph 3: yes – even

B1

4

(b) Graph 1: 8 edges

for considering repeats on 1 or 2

allow listings of CP

M1/B1

Graph 2: $10\text{ edges} + \frac{2}{\underline{12}}\text{ repeats}(BC\text{ and }FE)$

B1

Graph 3: $12\text{ edges} + \frac{0}{\underline{12}}\text{ repeats}$

B1

4

[8]

Q22.

(a) 1) *AC* 6

SCA (for MST)

M1

2) *AE* 7

Second

A1

3) *EG* 7

8 edges

M1

4) *GD* 1

5) *DB* 4

DB fifth

A1

6) *GI* 6

7) *IH* 7

8) *FC* 8

All correct

A1

Length = 46

[S.C. Using Kruskal's/no working, maximum 3/6.]

B1

6

(b) (i) Odd vertices *A* & *I*

May be implied

M1

Min connector is

ABDGI = 19

Route / Distance

A1

Total length = 19 + 121

Total + their repeat

M1

(= 140)

Tour, 21 vertices
(3*A*, 3*B*, 2*C*, 2*D*, 3*E*, 1*F*, 3*G*, 2*H*, 2*I*)

B1

4

(ii) Length = 140

B1

1

[11]

Q23.

(a) (i) Odd vertices ($\Rightarrow$ Repeats)

Eularian not enough

E1

1

(ii) Odd *BCDE*

considering odds

M1

BC + *DE* = 14
BD + *CE* = 12.5 + 10.5 = 23
<u>*BE*</u> + <u>*CD*</u> = 9 + 4.5 = <u>13.5</u>

A2,1,0

Distance = 13.5 + 82
= 95.5

82 + their (mir)

M1

A1F

Route: *A B C D E F B E B D C A*

or equivalent

B1

6

(b) (i) 5 × 3 = 15

B1

1

(ii) (n – 1)(n – 3)(n – 5)...× 1

M1

(n – 1)(n – 3)

A1

2

[10]

Q24.

(a) Odd vertices

E1

1

(b) ABE = 26
ACE = 20 + 2x
ADE = 25.5
26 ≥ 20 + 2x > 25.5

inequality with 20 + 2x

M1

$3 \geq x > 2.75$

or in words; allow $x < 3$
s.c: $x = 3$, x −= 2.75 scores 1 out of 2

A1

2

(c) Min. Dist = 185.5 + 3x + 25.5

for 185.5 + 3x

M1

= 211 + 3x

A1

Route must repeat *ADE*
e.g. *A B C A D C E B F E F D E D A*

15 vertices: 3A, 3D, 3E, 2B, 2C, 2F

B1

3

[6]

Q25.

(a) Total dist 25 × 4 = 100
+ (20 × 3) = 60

M1

+ 20 = 20

A1

(BE) 180
ABEBCDEFA

B1

3

(b) (i) B, E, P, Q, R, S

B1 for 4 correct

B2

2

(ii) Tour repeating
PQ, RS, EB

B1

Total 50 × 2 = 100
20 × 3 = 60

SCA

M1

Circle = 10π
Repeat 20 + 10 + 10 = 40
200 + 10π

AWRT 231.4

A1cao

3

[8]

Q26.

(a) *AB, CD*

AC, BD

AD, BC

B1 for 2 pairs

B2

2

(b) *AB* + *CD* = 8 + 9 =17

For using 9

B1

$$\left.\begin{array}{l} AB + BD = 5 + 5.5 + 10.5 \\ AD + BC = 4 + 7 = 11 \end{array}\right\}$$

For both

B1

Length = 10.5 +39.5

= 50

B1F

3

(c) 2

B1

1

[6]

Q27.

(a) (i) GB, WS
GW, BS

GS, BW

B1 **1**

(ii) GB, WS = 12 + 10 = 22
GW, BS = 9.5 + 13 = 22.5
GS, BW = 8.5 + 11 = 19.5
A1 for 1 correct pair

M1

A2

Total = 73.5 + 19.5 = 93

B1 **4**

(b) $5 \times 3 \times 1 = 15$

M1A1 **2**

(c) $(n - 1) \times (n - 3) \times (n - 5) \times \ldots\ldots \times 1$
A1 for $(n - 3)$ *A1 for 1*

M1A2 **3**

[10]

Q28.

(a) (i) RISCRWICWCR
or equiv.

M1 **2**

Must have RCW reference

A1

(ii) 1000

B1 **1**

(b) (i) RW makes odd

B1 **2**

vertices even

E1

(ii) $l < 80 + 80$

inequality (implied)

M1

2

I = 159

SC/ 160 B

A1

[7]

Q29.

(a) (i) C - D - E - F

any route between C and F

M1

C - D - E - F offered as the shortest route

A1

2

(ii) eg *ABCDEFEDCEBFA*

Route starting and finishing at A, with C to F repeated

M1

Complete correct route

ft route C to F in (i)

A1F

2

(iii) Network redrawn with two directed edges BE

Edges BE must be directed, one each way

B1

eg *ABEBCDECBFEFA*

A route, starting and finishing at A, with BE and EB both included

M1

Complete correct route, this time with BC and EF repeated

A1

3

(b) The start and finish need not be the same

E1 for any two (or equivalent) comments

E1

There are exactly two odd vertices, C and F.

E1 for the third (or equivalent) comment

E1

Start at C (or F) and finish at F (or C).

Alternative
give the complete route eg FABFEBCEDC

(E2)

2

[9]

Q30.

(a) Odd vertices implies

E1

edges have to be repeated

E1

2

(b) BD FG BF DG BG DF

any correct pair

M1A1

2

(c) 3 + 2.5 = 5.5
0.75 + 1.75 = 2.5

M1

any correct pair

A1

3.25 + 2.25 = 5.5

for 1.75 with 3 pairs listed

a1

3

(d) Route using BF DG
ABCDEFBFGEDGECA

M1

or equivalent

A1

Length 22

B1

3

[10]

Q31.

(a) (i) ***P S T...***

M1

... *Q U W...*

A1

... *R V P*

A1

Total 3 + 4 + 3 + 5 + 3 + 5 + 3 + 4 = 30 miles

B1

4

(ii) e.g. *QT*, *RV*, *UW*

M1

TS

A1

QU, *RW*

A1

Total 3 + 3 + 3 + 4 + 5 + 5 = 23 miles

B1

4

(iii) Hamiltonian route ≥

M1

Min conn + lowest two links to P
= 23 + 3 + 4 = 30.

A1

So the 30 found in (a) is best possible.

A1

3

(b) (i) The graph is K_8 with eight vertices of odd

M1

This needs at least four edges to
make it Eulerian.

A1

2

(ii) In order to add only 12 miles of roads

M1

look at the 4 roads of length 3 miles;
$PS \quad QT \quad RV \quad UW$.

A1

These do pair off the eight odd vertices and so repeating these roads will create an Eulerian graph.

A1

3

[16]

Q32.

(a) (i) There are vertices of odd degree, so the network is not Eulerian.

B1

1

(ii) Odd vertices: $A\ B\ F\ H$

M1

Pairings $AB\ FH$: adds 80+60

A1

$AF\ BH$: adds >>140
$AH\ BF$: adds >>140

A1

So shortest route has length 1100 + 140 = 1240 metres.

B1

4

(iii) e.g. $AHFDBACGHFBCEGFEBA$

M1A1

A1

3

(b) (i) $ACEGHFDBA$

M1A1

50 + 50 + 50 + 50 + 60 + 70 + 60 + 80 = 470

A1 3

(ii) AH + seven other arcs each ≥ 50

B1 1

(iii) Only two footpaths out of A are AB, AC. Similarly for D and H.

B1

Then the only way to complete the route via E is with CE, EG.

B1 2

(iv) (ii) the route of 470 beats any involving AH.

M

By (iii) the route of 470 is the *only* one not using AH and hence is the shortest.

A1 2

[16]

Q33.

(a) (i) Six odd vertices $PRSTUW$

Just B1 for "odd vertices"

B1

Pairing them off take at least 3 tracks

B1 2

(ii) Want to pair off PRSTUW to include PS

(or use any sensible short-cuts)

M!

or PU:

$PS\ RT\ UW$ $PU\ RS\ TW$

A1

$PS\ RU\ TW$ $PU\ RT\ SW$

$PS\ RW\ TU$ $PU\ RW\ ST$

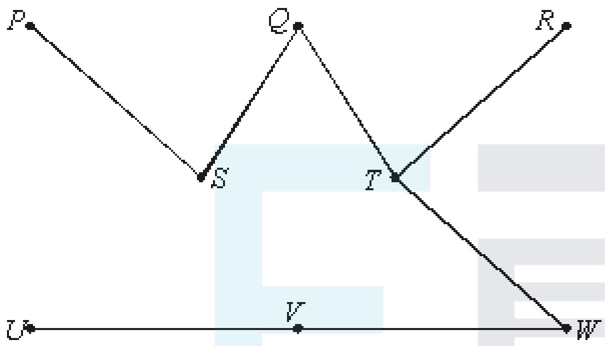
A1

Repeating $PS\ RT\ UV/VW$ takes 4 + 4 + 4 and is clearly unbeatable as each joined pair adds at least 4.

A1

5

(b) $UV\ VW\ QS\ PS\ QT\ RT\ WT$: 23



M1 A1

M1 A1

4

(c) (i) Trainspotter's cycle length ≥

25 + 25 + minimum connector length

= 50 + 23 = 73

M1

(ii) 73 not possible because (unique) minimum connector is not a path.

A1

Switching QT to QR makes it a path 1 mile longer, giving shortest round route:

B1 M1

e.g. Home $PSQRTWVU$ Home

A1

5

[16]

Q34.

(a) (i) Nearest neighbour approach gives

M1

$A \quad E \quad B \quad C \quad F \quad G \quad H \quad D \quad A$

A1A1 **3**

(ii) It uses **all** the 1p links

B1 **1**

(b) (i) Odd vertices $A \quad B \quad C \quad G$

M1

Pairings $ABCG$: 2 + 2
$AC \quad BG$: (2 + 1) + 2

A1

$AG \quad BC$; (2 + 2) + 1

A1

So repeat AB and CG

A1 **4**

(ii) Repeat BC (=1),

B1

with message starting at A and finishing at G.

B1 **2**

[10]

Q35.

(a) (i) There are some odd vertices

B1 **1**

(ii) Odd vertices A, E, F, G

M1

Pairings
AE/FG 12 + 10
AF/EG 10 + 11

A1

AG/EF 13 + 9

A1

Minimum number 21

B1

e.g. $ABFGEFBADGEBCD$

M1

ECA

A1

6

(b) (i) e.g. a "nearest neighbour" approach gives $ABFEDCA$

M1

with total 5 + 5 + 9 + 6 + 6 + 8 = 41 but

A1

with G missed out. Could detour EGD to include G making total 50

M1

(Any sensible approach for Hamiltonian or non-Hamiltonian route of length ≤ 50 is acceptable)

A1

4

[11]

Q36.

(a) Odd vertices $PQUX$

M1

Pairs PQ/UX, PU/QX, PX/QU

Adding 80, 100, 50 respectively

A1

Best is PX/QU giving total distance 345 + 50 + = 395

B1

e.g. $PQRVWSPTUQUVXWTXTP$

M1 A1

5

(b) Now 8 odd vertices

B1

So 105 pairings

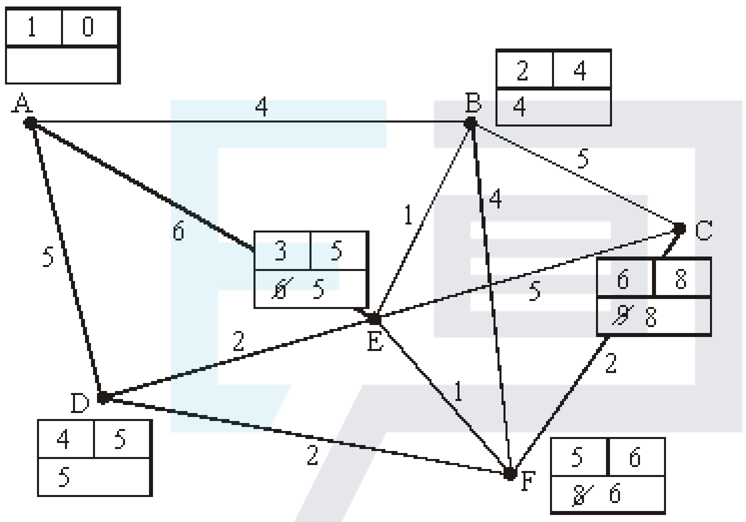
B1

2

[7]

Q37.

(a)



M1 Dijkstra
A1 Working values at E
A1 working values
A1 labels

M1A1A1A1

Repeat (uptake working value → label); then traceback B1

ABEFC 8 B1 B1 7

(b) AB 4

AD 5

ABE 5

ABEF 6

M1 using their (a)
A1 their ABEF
A1 their ABE
<u>ALT</u>
B1: ABE
B1: ABEF
B1: 5 & 6

M1A1A1

3

(c) eg. ADABCFECFBEFDEA; Dist = 45 M1A1A1A1

(Pairings: AC/DE AD/CE AE/CD
8+2 =10 5+3=8 5+4=9

M1 sca at route inspection

A1 DA/AD repeated

A1 pairing of odd nodes

A1 length

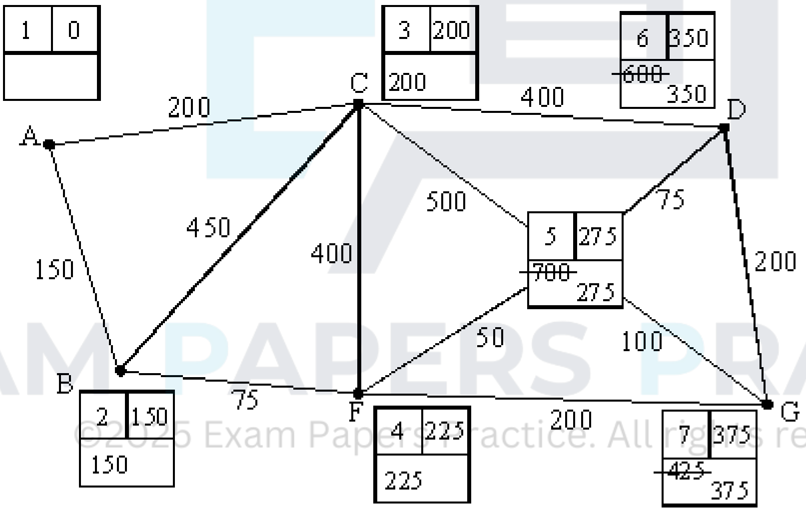
M1A1A1A1

4

[14]

Q38.

(a)



Dijkstra

M1

working values

A1

Labels

A1

Shortest route (traceback): ABFEG, with length 375

B1 B1

5

(b) odd vertices

B1

BC/DG: 350 + 175 = 525

B1

BD/CG: 200, 550

B1

BG/CD: 225, 400

B1

e.g. ACABCDEDGEGFECFBA

B1

Length = 2800 + 525 = 3325

B1

6

[11]

Q39.

(a) Odd vertice4s – A,C,D,G

M1

Repeat ABEC and DFG
can be implied

A2

2300 + 300 + 200 = 2800

A1

(b) 2×2300 = 4600 – all even nodes so transversal

M1 A1

Explanation

B1

(c) Digraph needed, but still transversal

B1

Route or argument such as “ins” = “outs” at each vertex

B1

[9]

Examiner reports

Q2.

Generally the answers to part (a) were a very high scoring. Working was shown clearly and calculated accurately although there was, as in previous papers, a surprising number of students who didn't believe that the total length of the network was 118! The second part caused more difficulties. Many students did not realise that 'the number of visits' = 'degree of vertex' / 2, and resorted to finding a detailed route, with varying degrees of success, and then counting each occurrence of a vertex.

Q3.

The method required for (a) was widely known, clearly explained and accurately calculated. The most common errors were to identify three or five odd order nodes. The incorrect answer of 3 was a frequently seen answer to part (b), obtained by dividing 6 by 2 and ignoring the correct method applied in the previous part.

Q4.

This question wasn't answered very well, although zero scores were very rare. Candidates knew the required method but less than half were able to calculate the required lengths correctly in part (a). Part (b) was no better, 2 being the popular answer.

Q5.

Part (a) was well-answered – students had been well-drilled in the rote use of an elementary algorithm, although arithmetical accuracy was sometimes weak. However, in the remainder of the question, students were required to do a little elementary analysis and thinking and many failed completely. It appears that these modified situations are not handled well. Students need to spend time studying semi-Eulerian situations.

Q6.

Parts (a) and (b) were answered well. The only significant error was to calculate the value of BG incorrectly - the least length being 210, not 225.

Part (c) did cause problems, usually part (c)(i) where 2 was the common wrong answer.

Q7.

This question clearly differentiated between those candidates who could follow an algorithm and those who understood the method fully. In part (a), many candidates made mistakes on the pairings but overall this part was well attempted. However, mistakes were made when, after finding the correct odd nodes, the distances between them were incorrectly calculated. Almost all found the correct minimum pair sum, $AC + FD = 32$, and thus the correct final answer. Full credit for this was not allowed unless all the pair sums had been correctly calculated. Parts (b) and (c) were not well answered. There was much confusion as to what might be added to, or subtracted from, 150 or 180.

Q8.

Most candidates scored full marks or, at worst, dropped 1 mark: $BD + CE$ in part (b).

Q9.

Part (a) The method was widely known and the correct 4 odd nodes usually identified, as were the required three pairs. Finding the correct lengths proved much more difficult and many marks were lost there.

Part (b) This was poorly answered, with 8 being the most common answer.

Q10.

This question was pleasingly well done, even by those candidates who had not managed part (a) correctly.

Q11.

In part (a), the method was well known and most candidates scored heavily, although a surprising number failed to get the correct total for all three pairs.

After that there were a lot of problems and the question discriminated well. Roughly a third of candidates thought that parts (b) and (c)(i) could be solved by inspection and about half of them were successful. Another third resorted to attempting to work out the route, apparently by trial and error, and were usually unsuccessful. The remainder either omitted the questions or wrote a random number down. The clear majority answer for part (c)(ii) was A, B, C and D. Once again, the able candidates were usually correct.

Q12.

The majority of candidates correctly answered part (a). In part (b) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting the odd vertices. Candidates must realise that they must consider the shortest distances connecting vertices when solving a Chinese postman problem. Many candidates who obtained the correct minimum pairing AI + BC failed to obtain the correct answer of 500.

Part (c) was poorly answered. Candidates should realise that the purpose of pairing odd vertices is to add extra edges to the network. This makes the network Eulerian and it is immediately apparent as to the number of times each vertex would appear in an optimum route.

Q13.

This question proved to be a good discriminator. In part (a) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting the odd vertices. Candidates must realise that they must consider the shortest distances connecting vertices when solving a Chinese postman problem. Parts (b) and (c) were poorly answered. Many candidates produced what appeared to be a random selection of numbers, with little thought as to the requirements of the question. These responses showed a lack of an in-depth understanding of the theory relating to Chinese postman problems.

Q14.

Nearly all candidates were able to answer part (a) correctly. In part (b), candidates must realise that for a Chinese postman problem involving 4 odd vertices, they have to show the 3 pairings **and** justify that they have selected the optimum pairing. A number of candidates thought that the sum of the 3 pairings was the same and that any pairing could be chosen. Some candidates who correctly chose the correct pairing to repeat failed to find a corresponding route. Candidates are encouraged to list the orders of the vertices after the repeats have been included. They would then be able to check if they had a correct route corresponding to their minimum distance.

Q15.

In this question all candidates were able to score marks, although full marks were rarely seen. Candidates should realise that for a Chinese postman problem involving four odd vertices, they have to show the three pairings AND justify their choice for the optimum pairing.

Q16.

Candidates scored highly on this question, although full marks were obtained by less than half of the entry. Candidates needed to indicate that a route from F to D had to be repeated and that the direct road, of length x miles, was the shortest route.

Q17.

The majority of candidates performed well in part (a), although some weaker candidates thought that the minimum spanning tree should produce a full cycle.

Part (b) was poorly answered. Many candidates thought that this part was independent of part (a) and merely found the length of a Chinese Postman route for the four vertices mentioned in the table. Although this approach was fundamentally wrong, candidates still received some credit for their work on the odd vertices.

Q18.

The majority of candidates correctly answered part (a)(i). In part (a)(ii) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting $CD+EF$ and $CF+DE$. Candidates should realise that they must consider the shortest paths connecting vertices when solving a Chinese postman problem. Many candidates who obtained the correct minimum of 350 by pairing $CD+EF$ failed to use this fact and repeated the direct connection of CD. Part (b) was poorly answered. It is a standard piece of bookwork that candidates should know.

Q19.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used 16 + 2 or 3. In part (ii), a common mistake was to omit the edge EL. Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing AE and LP, then E and L would become the odd vertices.

Q20.

This question proved to be a good discriminator. Mistakes were made in the third part of the question. After substituting $x = 10$, candidates made the wrong choice as to the route to be repeated. The final part was only answered well by the best candidates. To gain full marks candidates had to show that for any value of x the route $B\ C\ D\ E$ should never be chosen. A significant number of candidates substituted random values for x and thought this was a proof.

Q21.

This question was well answered with many candidates scoring the full 8 marks. In the first graph in part (a) many candidates gave the answer of semi-Eulerian which was acceptable and in part (b) a number of candidates listed the Chinese postman route when all that was required was the length. These candidates were not penalised.

Q22.

Candidates scored high marks on part (a) of this question, although there was a significant number of scripts with careless slips. In part (b) candidates realised that the length of an optimal Chinese Postman route must include a repetition of A to I but did not use the minimum spanning tree found in part (a) to find the minimum additional distance.

Q23.

Although candidates realised that an optimal 'Chinese postman' route depended on odd vertices, there was a disappointing number who failed to pair off the odd nodes correctly to arrive at the correct minimum repetition. A surprising number of candidates who did arrive at the correct numerical answer failed to write down a route. Part (b) proved to be the most difficult part of the paper with only a few candidates obtaining the correct answer for part (b)(ii).

Q24.

Many candidates knew that 'odd' had something to do with part (a) and then elaborated vaguely on this word - they were not penalised for this. Most candidates were able to attempt part (b) but many displayed poor arithmetic skills – for example, it was common to see 5.5 ÷ 2 = 2.25 Some candidates did not see the connection between parts (b) and (c) and they included a repeat of ACE in their total.

Q25.

Candidates were expected to find, and pair, odd vertices to find an optimal solution. The question was answered well with the majority of candidates scoring full marks.

Q26.

This question was answered well by the majority of candidates, with many scoring full marks.

Q27.

Candidates were able to start the question and able to pair the odd vertices but they found difficulty in finding the shortest pairing. Part (b) was correctly answered by a few of the candidates but there were no fully correct answers to part (c).

Q28.

The question was well answered although a number of candidates did not realise that the quickest route from R to W was via C. Some candidates did not read part (b)(ii) carefully as they failed to give their answer as an integer value.

Q29.

The candidate was able to work with the odd vertices but didn't consider which was the shortest route to connect these two vertices. The same difficulty arose in answering part (a)(iii). Part (b) was correctly answered.

Q30.

Parts (a) and (b) were well answered but candidates assumed, wrongly, that the shortest distance between the odd vertices would be the direct route, leading to numerical errors in part (c). There were very few correct answers to part (d).

Q34.

Part (a) was well done but, surprisingly, the majority of candidates seemed to struggle with the Chinese postperson problem in part (b).

Q36.

In part (a), some candidates were a little slip-shod in their application of the Chinese postperson algorithm, not showing clearly that they had considered all possible pairings of the four odd vertices.

Q38.

In part (a) of this question evidence of the required algorithmic work was often lacking. Only a minority of candidates systematically considered all possible pairings of odd vertices in part (b), as required.

Q39.

Most candidates attempted this question, which discriminated well.

In part (a), a surprisingly large proportion of candidates did not quote the name (or provide any evidence) of the “appropriate algorithm” required (route inspection). Many did not appear to have considered odd vertices at all; there was little evidence of computing distances to obtain the optimal pairing. Some candidates noted that AB must start and BA end the route. They eliminated the arc AB, making B odd. This was a perfectly valid simplification.

In part (b) it was intended that candidates would realise that the nodes were all even, so that the route was traversable and twice the original length. Some candidates produced a correct route and corresponding correct distance, added up arc by arc. Demonstrated calculation of the distance was accepted as a valid justification for the answer, although it was far from elegant.

Many candidates realised that the distance in part (c) was the same as in part (b). Some were able to justify this conclusion, obtaining the final mark, by giving their route correctly. There were some attempts at the (difficult) justification of the result in general.